Tasfia Zahin

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https://tasfiazahin.github.io/

Education

The Pennsylvania State University — University Park, PA, USA

Ph.D. in Computer Science and Engineering — Aug 2021 - May 2026

- Advisor: Dr. Mingfu Shao
- Thesis: Computational Approaches for Full-Length Circular RNA Assembly from RNA Sequencing Data

Bangladesh University of Engineering and Technology (BUET) — Dhaka, Bangladesh

Bachelor's in Computer Science and Engineering — Feb 2015 - Apr 2019

- CGPA: 3.90/4.00

Work Experience

NSF National Synthesis Center for Emergence in the Molecular and Cellular Sciences (NCEMS), Huck Institutes of the Life Sciences — University Park, PA, USA

Postdoctoral Scholar — Aug 2026 - Present

- Project Lead - Epigenetic Drivers: Genome Structure-Function Relationships in Human Brain Cells

The Pennsylvania State University — University Park, PA, USA

Graduate Research/Teaching Assistant — Aug 2021 - June 2026

Ahsanullah University of Science and Technology (AUST) — Dhaka, Bangladesh

Lecturer, Computer Science and Engineering — July 2019 - July 2021

Dhanmondi Tutorial — Dhaka, Bangladesh

Teacher — Jan 2019 - April 2019

Publications

1. **Tasfia Zahin**, Irtesam Mahmud Khan, Mingfu Shao. Accurate Reconstruction of Circular RNAs from Complex Rolling Circular Long Reads with CircPlex. *bioRxiv*, Nov 2025. https://doi.org/10.1101/2025.11.21.689841
2. Zhezheng Song*, **Tasfia Zahin***, Xiang Li, Mingfu Shao. Accurate Detection of Tandem Repeats from Error-Prone Sequences with EquiRep. *Genome Research*, December 2025 35: 2714-2721.
3. Zhezheng Song*, **Tasfia Zahin***, Xiang Li, Mingfu Shao. Accurate Detection of Tandem Repeats from Error-Prone Sequences with EquiRep. Proceedings of 29th *International Conference on Computational Molecular Biology (RECOMB 2025). Lecture Notes in Computer Science*, vol 15647, pp. 390–394. Springer, Cham, 2025 (extended abstract).
4. **Tasfia Zahin**, Qian Shi*, Xiaofei Carl Zang*, Mingfu Shao. Accurate Assembly of Circular RNAs with TERRACE. *Genome Research*, September 2024 34: 1365-1370.
5. **Tasfia Zahin**, Qian Shi*, Xiaofei Carl Zang*, Mingfu Shao. Accurate Assembly of Circular RNAs with TERRACE. Proceedings of 28th *International Conference on Computational Molecular Biology (RECOMB 2024). Lecture Notes in Computer Science*, vol 14758, pp. 444–447. Springer, Cham, 2024 (extended abstract).
6. Anubhav Sen, **Tasfia Zahin**, Mingfu Shao. Chimeric and Outward Reads in Short-Read RNA Sequencing Data: A Unified Circular RNA Assembly Framework. Under review at RECOMB 2027.
7. Xiaofei Carl Zang, **Tasfia Zahin**, Irtesam Mahmud Khan, Qian Shi, Yi Xing, Mingfu Shao. Amaranth: Enhanced Single-Cell Transcript Assembly via Discriminative Modeling of UMI Reads and Internal Reads. *Bioinformatics*, Volume 42, Issue Supplement_1, July 2026, btag292. Proceedings of 34th *Conference on Intelligent Systems for Molecular Biology (ISMB 2026).*

8. Irtesam Mahmud Khan, Xiaofei Carl Zang, Ange Teng, **Tasfia Zahin**, Mingfu Shao. Boosting Transcript Assembly via Delineating Transcript Start and End Sites. *bioRxiv*, Oct 2025. `https://doi.org/10.1101/2025.10.13.682211`
9. **Tasfia Zahin***, Md Hasin Abrar*, Mizanur Rahman Jewel, Tahrina Tasnim, Md Shamsuzzoha Bayzid, Atif Rahman. An alignment-free method for phylogeny estimation using maximum likelihood. *BMC Bioinformatics*, 2025 Mar 7;26(1):77.

Academic Services

- Sub-reviewer - *WABI 2026*, *ISMB 2026*, *RECOMB 2026*, *RECOMB 2025*, *WABI 2025*, *RECOMB 2024*, *ISMB 2024*, *WABI 2024*, *WABI 2023*, *RECOMB 2022*, *ACM-BCB 2022*.
- Research Mentor - Schreyer Honors College, Penn State undergrads (2024-2026).
- Volunteer - The 2024 Workshop on Emerging Methods for Sequence Analysis (WEMSA 2024), The Pennsylvania State University, Pennsylvania, USA.

Conference Participation

- Trainee - NSF-NCEMS Annual Summit Meeting 2026
- Research and poster presenter - RECOMB 2025, Seoul, South Korea.
- Research presenter - RECOMB 2024, Boston, Massachusetts, USA.
- Volunteer - The 2024 Workshop on Emerging Methods for Sequence Analysis (WEMSA 2024), The Pennsylvania State University, Pennsylvania, USA.
- Research presenter - The 2023 Workshop on Emerging Methods for Sequence Analysis (WEMSA 2023), The Pennsylvania State University, Pennsylvania, USA.
- Attendee - ISMB 2022, Madison, Wisconsin, USA.
- Attendee - RECOMB 2020 Virtual Conference.

Teaching Experience

- Graduate Teaching Assistant - The Pennsylvania State University, Pennsylvania, USA.
  - CMPSC 465 Data Structures and Algorithms - Fall 2021, Fall 2024.
- Lecturer - Ahsanullah University of Science and Technology, Dhaka, Bangladesh.
  - Implemented Outcome-Based Education (OBE) principles in curriculum and course design - aligning course learning outcomes, instructional strategies, and assessment methods.
  - CSE 3107 Microprocessors and CSE 3108 Microprocessors Lab - Spring 2019, Fall 2019.
  - CSE 1287 Computer Programming and CSE 1288 Computer Programming Lab - Spring 2020.
  - CSE 2106 Digital Logic Design Lab - Fall 2020.
  - CSE 4126 Distributed Database Systems Lab - Spring 2019.
  - CSE 2186 Basic Programming Techniques Lab - Fall 2020, Fall 2019.
  - CSE 2146 Introduction to Computer Science - Spring 2020.

Professional and Student Engagement

- Anchor/MC and IT support sub-committee member - The 23rd International Conference on Computer and Information Technology (ICCIT) 2020, Ahsanullah University of Science and Technology.
- Member of executive committee - Robotics and Innovation Club, Ahsanullah University of Science and Technology, 2020-2021.
- Member of evaluation committee - Intra-Department Project Show, Ahsanullah University of Science and Technology, 2020.
- Mentor - Student Programming Development Program, Ahsanullah University of Science and Technology, 2020-2021.

- Mentor - Bangladeshi Women in Computer Science and Engineering (BWCSE) Mentor-mentee program, Bangladesh University of Engineering and Technology, 2020-2021.
- Media and publication secretary - English Medium Buetians' Association (EMBA), Bangladesh University of Engineering and Technology, 2018-2019.
- Member of organization committee, CSE Festival 2019, Bangladesh University of Engineering and Technology.
- Adjudicator - BRAC Inter-University Debate Competition 2016.
- Delegate - Bangladesh Youth Leadership Summit 2016.
- Member - Bangladeshi Women in Computer Science and Engineering (BWCSE), Bangladesh University of Engineering and Technology, 2015-2019.
- Member - Debating Club, Bangladesh University of Engineering and Technology, 2015-2017.

Awards and Honors

- Winner, Huawei Seeds for the Future 2019, Travel Grant by Huawei Technologies.
- Dean's List Honor, 2015-2019, Bangladesh University of Engineering and Technology.
- Daily Star O and A Level Award of Excellence, 2012 and 2014, The Daily Star Newspaper, Bangladesh.